Journal of
Plant Ecology

Editorial

Bibliometric analysis of *Journal of Plant Ecology* during 2017–2021

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Journal of Plant Ecology (JPE) was founded in 2008. It is sponsored by the Botanical Society of China and the Institute of Botany, Chinese Academy of Sciences, and published by Oxford University Press, UK. JPE publishes diverse types of articles that fall into the broad scope of plant ecology, including plant ecophysiology, population ecology, community ecology, ecosystem ecology, landscape ecology, conservation ecology, evolutionary ecology, theoretical ecology and global change ecology.

JPE has transferred to fully Open Access since July 2021, making the full text freely available for readers and authors. JPE serves as an important medium for plant ecologists worldwide to publish their research findings. Here, we present a bibliometric analysis (Aria and Cuccurullo 2017) to visualize the publication statistics and the document network of research papers published in JPE during 2017–2021.

The raw dataset was downloaded from the Web of Science Core Collection on 18 October 2022, including the fields author, title, journal title, cited frequency, abstract, address, affiliation, publication type, author keywords, keywords plus, Web of Science category, research field, references and the number of references. The raw data were manually cleaned, duplicated and items not related to research (4 editorials, 3 corrigenda, 2 commentaries and 1 data paper) removed, resulting in 472 items (466 research articles and 6 review articles) that could be further analyzed. All analyses were conducted using R v.4.2.1 (R Core Team 2022).

Publication overview of JPE

During 2017–2021, JPE published an average of 96 articles annually and had an acceptance rate of 28.2% (Table 1). From the start of JPE in 2008 up to the study period, both the number of submissions and the number of published articles tripled (Schmid *et al.* 2018; Zhang *et al.* 2020c). In addition to the traditional research and review papers, JPE has also published data papers, method papers as well as perspective and commentary papers (Fig. 1).

Most productive authors, countries and institutions for papers published in JPE

During 2017–2021, a total of 1890 authors published their works in JPE. The 10 most prolific authors come from three different countries and six of them obtained more than 100 citations (Table 2a). Among all articles, 229 are from institutions in China, accounting for 48.5% of the total publications in the journal. These articles received 1365 citations (Table 2b).

The authors of articles published in JPE during 2017–2021 are from 618 institutions. The top 10 institutions in terms of number of publications are shown in Table 2c. The Chinese Academy of Sciences system is the most productive institution, which includes the University of Chinese Academy Sciences, Institute of Botany, Institute of Geographic Sciences and Natural Resources Research and Institute of Applied Ecology. It is then followed by Peking University and the University of Zurich (Table 2c).

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Table 1: Overview of published article and submission statistics of JPE during 2017–2021

Year	No. of issues	No. of published articles	No. of submissions	No. of accepted articles	Acceptance rate in %	Days to final decision
2017	6	98	214	51	23.8	85
2018	6	91	253	63	24.9	66
2019	6	101	250	61	24.4	70
2020	6	92	400	138	34.5	77
2021	6	97	342	118	34.5	72

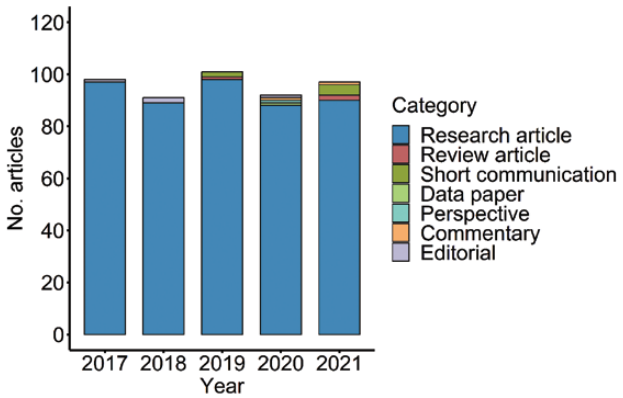

Figure 1: The annual publications in JPE during 2017–2021.

Most cited articles of JPE

The 10 most cited papers among the published papers in JPE during 2017–2021 and 2020–2021 are presented in Table 3. Among the first set, most papers were published in 2017, and two were published in 2018 (Gioria *et al.* 2018; Rodrigues *et al.* 2018) (Table 3a). In particular, the special issue on 'Biodiversity–Ecosystem Functioning' in 2017, including 24 research articles and 1 editorial, received high citations. Six articles from this special issue ranked among the 10 most highly cited papers from 2017–2021 (Bu *et al.* 2017; Huang *et al.* 2017; Li *et al.* 2017; Schmid *et al.* 2017; Scholten *et al.* 2017; Sun *et al.* 2017). To evaluate the performance of recently published papers in JPE, we also analyzed the 10 most highly cited articles within the past 2 years (2020–2021; Table 3b). Two articles published during this period are included in top 10 list for the entire period 2017–2021 with citation-per-year values of 7.33 (Li *et al.* 2020) and 6.50 (Liu *et al.* 2021), respectively.

Co-occurrence network of keywords

The co-occurrence network of KeyWords Plus in JPE was generated using the 'biblioNetwork' function in the 'bibliometrix' package (Aria and Cuccurullo 2017) (Fig. 2). The definition of KeyWords Plus by Web of Science is that 'KeyWords Plus are index terms automatically generated from the titles of cited articles. KeyWords Plus terms must appear more than once in the bibliography and are ordered from multi-word phrases to single terms. KeyWords Plus augments traditional keyword or title retrieval.' (Web of Science Core Collection Help; https://images.webofknowledge.com/images/help/WOS/hp_full_record.html). In the co-occurrence network, a total of 50 keywords are displayed based on the points' relative position (Singh *et al.* 2022) and grouped into four clusters. The red zone (Cluster 1) indicates the diversity theme, the green zone (Cluster 2) represents the climate-change theme, the purple zone (Cluster 3) shows the theme of evolution and the blue zone (Cluster 4) displays the theme of plants. Briefly, diversity and climate change are the main topics that appear mostly in research articles published in JPE during 2017–2021 (Fig. 2).

Conceptual structure map of keywords

A conceptual structure map of keywords was created based on multiple correspondences analysis (MCA). The conceptual structure map shows the relationship between one word and another by regional mapping, and each word is positioned according to the values of dimension 1 (Dim 1) and dimension 2 (Dim 2). MCA is used in bibliometric analysis, to develop a map between words with relatively similar values (Rahaman *et al.* 2022). The conceptual structure map can be used to identify clusters of documents that express common concepts (Aria and Cuccurullo 2017). A total of 50 words were grouped into two clusters on the map. The red zone (Cluster 1) represents the theme of climate change (how plants respond to environmental changes), and the blue zone (Cluster 2) indicates the theme of evolution (Fig.

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Table 2: Top contributors of articles in JPE during 2017–2021

Item	NP	*h*-index	*g*-index	TC
(a) Most productive authors (country)				
Schmid B (Switzerland)	13	11	13	384
Bruelheige H (Germany)	10	8	10	263
Yu FH (China)	9	6	8	67
Li Y (China)	8	5	8	120
Zhang Y (China)	8	3	5	27
Fang JY (China)	7	5	7	57
Ma KP (China)	7	4	7	115
Fischer M (Switzerland)	6	6	6	165
Niklaus PA (Switzerland)	6	6	6	267
Li W (China)	6	4	5	34
(b) Most productive countries				
China	229			1365
USA	43			200
Brazil	26			177
Germany	25			343
Spain	23			126
Australia	12			63
Canada	11			69
Czech Republic	8			130
Italy	8			84
Switzerland	8			170
(c) Most productive institutions				
University of Chinese Academy Sciences	72			
Chinese Academy of Sciences	53			
Peking University	51			
Institute of Botany, CAS	48			
University of Zurich	20			
German Center Integrative Biodiversity Research (iDiV)	18			
Institute of Geographic Sciences and Natural Resources Research, CAS	18			
Institute of Applied Ecology, CAS	15			
The University of Lavras	15			
Martin Luther University Halle-Wittenberg	14			

NP = number of publications, TC = total citation. *h*-index: a scientist has index *h* if *h* of his/her Np papers have at least *h* citations each, and the other (Np-*h*) papers have no more than *h* citations each (Hirsch 2005); *g*-index: given a set of articles ranked in decreasing order of the number of citations that they received, the *g*-index is the unique largest number such that the top *g* articles received together at least g^2 citations (Egghe 2006).

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Table 3: Top-cited articles published in JPE

TC	Title	Authors (year)	CPY
(a) Articles published during 2017–2021			
73	On the combined effect of soil fertility and topography on tree growth in subtropical forest ecosystems—a study from SE China	Scholten *et al.* (2017)	12.17[2]
62	A guide to analyzing biodiversity experiments	Schmid *et al.* (2017)	10.33[3]
61	Timing is everything: does early and late germination favor invasions by herbaceous alien plants?	Gioria *et al.* (2018)	12.20[1]
47	Spatial variations in responses of vegetation autumn phenology to climate change on the Tibetan Plateau	Cong *et al.* (2017)	7.83[4]
46	Positive effects of tree species diversity on litterfall quantity and quality along a secondary successional chronosequence in a subtropical forest	Huang *et al.* (2017)	7.67[5]
46	Positive effects of tree species richness on fine-root production in a subtropical forest in SE-China	Sun *et al.* (2017)	7.67[5]
38	The effect of tree size, neighborhood competition and environment on tree growth in an old-growth temperate forest	Zhang *et al.* (2017)	6.33[9]
35	Interspecific and intraspecific variation in specific root length drives aboveground biodiversity effects in young experimental forest stands	Bu *et al.* (2017)	5.83[10]
33	Crown and leaf traits as predictors of subtropical tree sapling growth rates	Li *et al.* (2017)	5.50
32	The influence of soil on vegetation structure and plant diversity in different tropical savannic and forest habitats	Rodrigues *et al.* (2018)	6.40[8]
(b) Articles published during 2020–2021			
22	Spatiotemporal variation in leaf size and shape in response to climate	Li *et al.* (2020)	7.33[6]
14	Species-specific responses to drought, salinity and their interactions in *Populus euphratica* and *P. pruinosa* seedlings	Yu *et al.* (2020)	4.67
14	Effects of warming on soil respiration during the non-growing seasons in a semiarid temperate steppe	Miao *et al.* (2020)	4.67
13	Sexual differences and sex ratios of dioecious plants under stressful environments	Liu *et al.* (2021)	6.50[7]
13	Contrasting responses of native and alien plant species to soil properties shed new light on the invasion of dune systems	Vitti *et al.* (2020)	4.33
11	The role of soluble sugars during drought in tropical tree seedlings with contrasting tolerances	O'Brien *et al.* (2020)	3.67
10	Interactive effect of climate warming and nitrogen deposition may shift the dynamics of native and invasive species	Ren *et al.* (2021)	5.00
10	Trans-generational effects in the clonal invader *Alternanthera philoxeroides*	Portela *et al.* (2020)	3.33
9	Congeneric invasive versus native plants utilize similar inorganic nitrogen forms but have disparate use efficiencies	Yu and He (2021)	4.50
9	The ultraviolet colour component enhances the attractiveness of red flowers of a bee-pollinated plant	Chen *et al.* (2020)	3.00

CPY = citation per year, TC = total citation. [1–10]Rank in order of CPY among the 20 articles.

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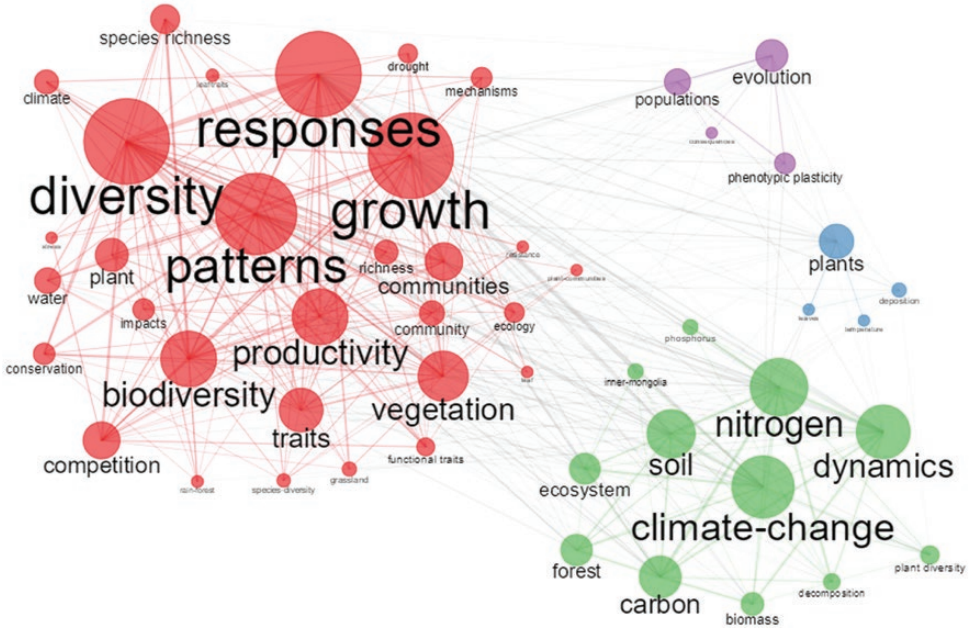

Figure 2: Co-occurrence network of keywords plus in JPE papers during 2017–2021.

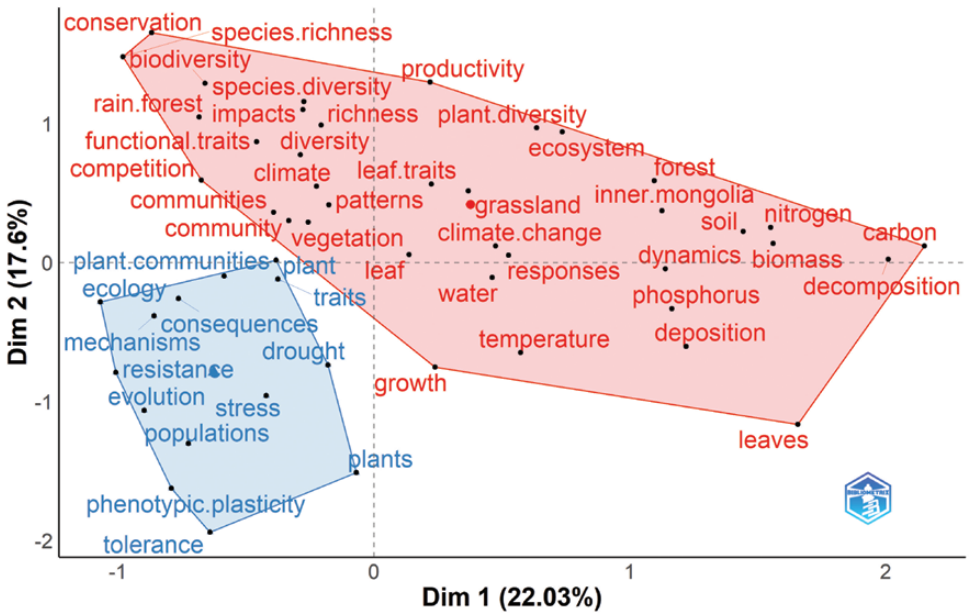

Figure 3: Conceptual structure map of keywords plus in the JPE papers during 2017–2021.

3). In addition, the red zone includes a higher number and variety of words, suggesting that many research articles published in JPE during 2017–2021 are relevant to these themes shown in this region (Fig. 3). Moreover, the documents that make the most contribution associated with the two clusters are shown in Table 4, and five papers are detected for each of the cluster. The five articles with high contributions in Cluster 1 focused on plant communities (Ochoa-Hueso *et al.* 2021; Wei *et al.* 2019; Xu *et al.* 2020; Zhang *et al.* 2020a, 2020b), and the other five articles most representative of Cluster 2 focused on specific plant species (Kirschbaum *et al.* 2021; Liu *et al.*

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Table 4: Papers with strongest associations with two main themes in JPE during 2017–2021

Title	Author (year)	Contributes
Cluster 1: climate change theme		
Experimental evidence for weakened tree nutrient use and resorption efficiencies under severe drought in a subtropical monsoon forest	Xu *et al.* (2020)	4.16
Root production, mortality and turnover in soil profiles as affected by clipping in a temperate grassland on the Loess Plateau	Wei *et al.* (2019)	4.16
Understorey plant community assemblage of Australian *Eucalyptus* woodlands under elevated CO_2 is modulated by water and phosphorus availability	Ochoa-Hueso *et al.* (2021)	3.81
Fluorescence characterization and microbial degradation of dissolved organic matter leached from salt marsh plants in the Yellow River Delta	Zhang *et al.* (2020a)	3.19
Characteristics of fungal community structure during the decomposition of mixed foliage litter from *Pinus massoniana* and broadleaved tree species in southwestern China	Zhang *et al.* (2020b)	3.15
Cluster 2: evolution theme		
Variation in regrowth ability in relation to land-use intensity in three common grassland herbs	Kirschbaum *et al.* (2021)	3.97
Species-specific effects of genetic diversity and species diversity of experimental communities on early tree performance	Zeng *et al.* (2017)	2.70
Linking plant spatial aggregation with reproductive traits and near-source seed dispersal: ecological adaptation to heavy grazing	Liu *et al.* (2020)	2.11
Congeneric invasive versus native plants utilize similar inorganic nitrogen forms but have disparate use efficiencies	Yu and He (2021)	1.76
Invasive plant rhizome production and competitiveness vary based on neighbor identity	Smith *et al.* (2021)	1.66

2020; Smith *et al.* 2021; Yu and He 2021; Zeng *et al.* 2017).

Thematic analysis through bibliographic coupling

The thematic map of keywords is created with 250 words being grouped into 17 clusters. The sizes of the bubbles on the map depend on the number of publications in which the keywords appear (Rejeb *et al.* 2022). Themes with high density and high centrality, appearing in the first quadrant and namely motor themes, have strong internal and external ties, which are important and well developed. Themes with high density and low centrality, namely niche themes, have well-developed internal ties and marginally significant external ties, which appear in the second quadrant. Themes with low density and centrality in the third quadrant, namely *emerging-or-declining themes*, have both weak internal and external ties. Themes with low density and high centrality appearing in the fourth quadrant, namely *basic themes*, suggest well-developed external ties and unimportant internal ties. As shown in the map (Fig. 4), the themes including the keywords of rain-forest, leaf traits and leaf area are important and well developed in JPE during 2017–2021.

In conclusion, as a relatively young journal dedicated to all fields of plant ecology, JPE has established distinct features covering topics from microcosm to macrocosm, with diversity and climate change as the most frequently appearing key words. We encourage authors to submit their manuscripts from widely ranging topics of classical and contemporary ecology to JPE. We thank the authors, editors and readers for their consistent support to JPE, and we continue to aim building JPE as an internationally recognized journal with global impact.

Conflict of interest statement. The authors declare that they have no conflict of interest.

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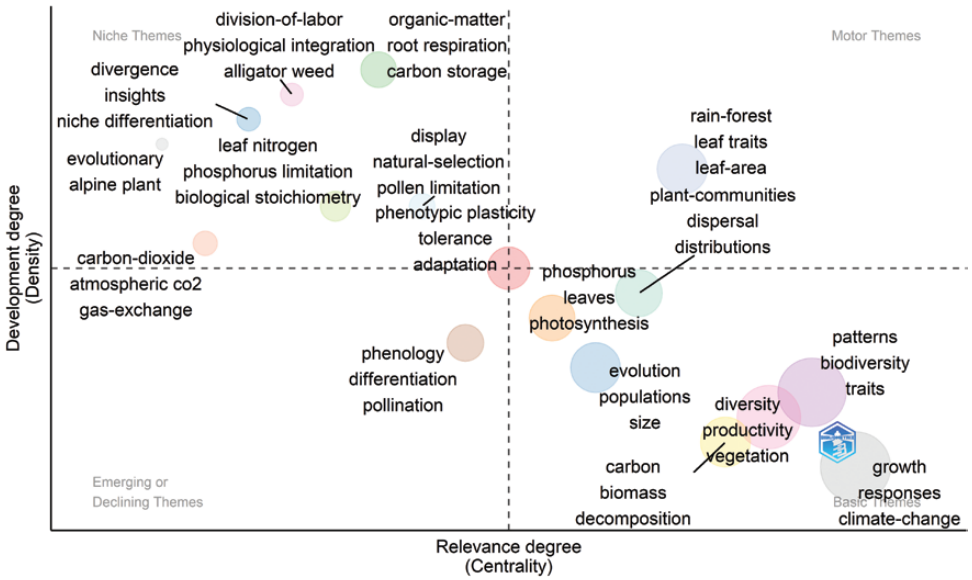

Figure 4: Thematic map of keywords plus in the JPE papers during 2017–2021.

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